You can perform validation on form data in two places in a web application: in the browser, using either client-side script or the browser's in-built data type validation, and on the server, using C# code. However, you should only ever view client-side validation as a courtesy to the user because it is easily circumnavigated by anyone who knows how to use the browser's developer tools, for example. Server-side validation should be seen as essential.

The MVC framework, on which Razor Pages is built, includes a robust validation framework that works against inbound model properties on the client side and the server. This framework massively reduces the developer burden of writing validation code and maintaining it in two places.

The key players in the input validation framework are

- `DataAnnotation` attributes
- jQuery Unobtrusive Validation
- Tag helpers
- `ModelState`

## *5.3.1 DataAnnotation attributes*

The primary building block of the validation framework is a set of attributes that inherit from `ValidationAttribute`. Most of these attributes reside in the `System.ComponentModel.DataAnnotations` namespace. Each attribute is designed to perform a specific type of validation—be it for presence, data type, or range. Some also enable you to test the incoming value against an expected pattern. Table 5.2 lists the validation attributes you are most likely to use and the type of validation they provide along with example usage.